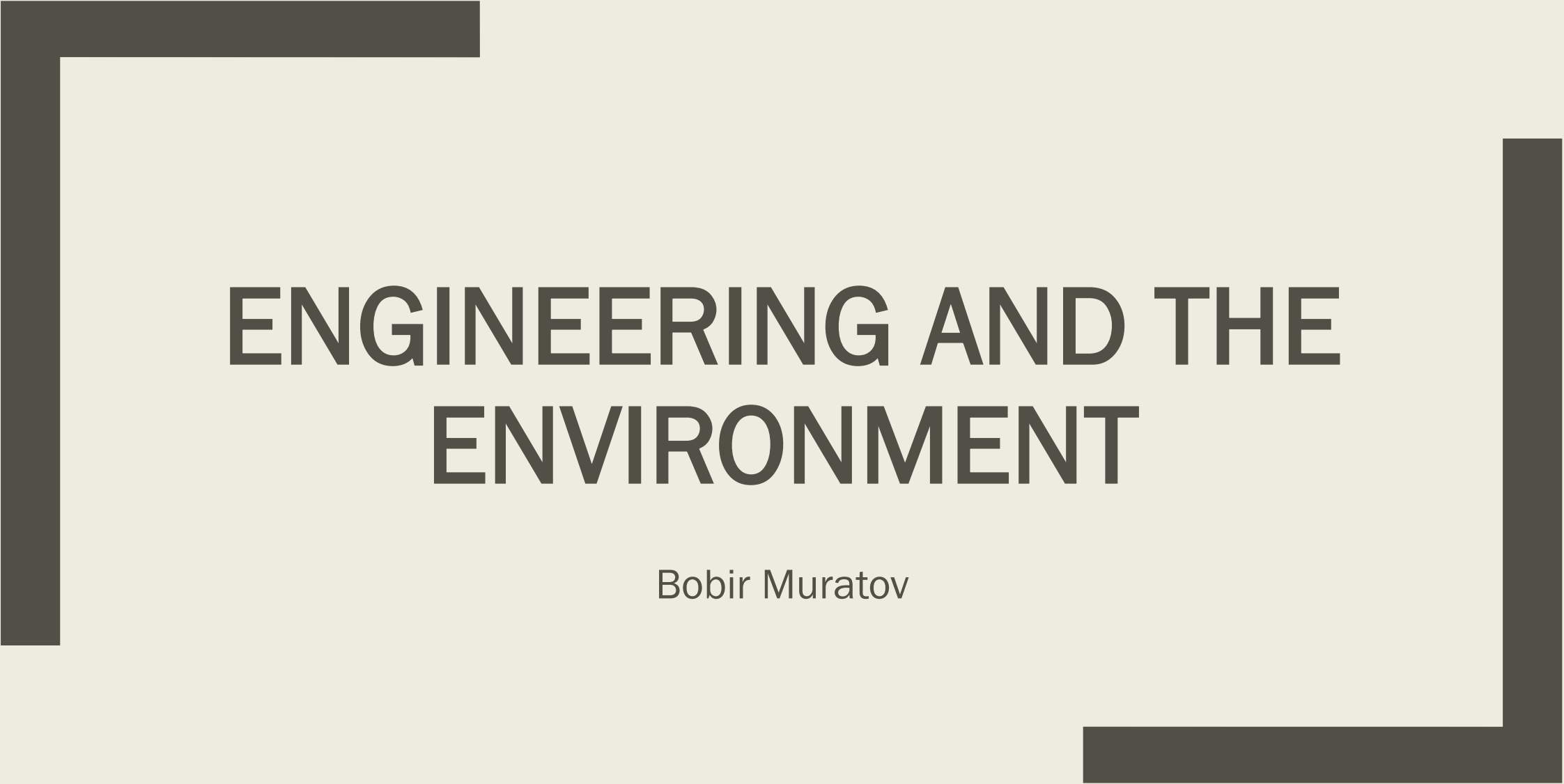


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ENGINEERING AND THE ENVIRONMENT

Bobir Muratov

Case

Prof. Bryan Willson



Introduction

- Responsibility to promote human well-being
- Impact of technology on the environment

Development of the Modern Environmental Movement

- How and when the modern environmental movement began?
- Can you give examples from the foundations, history, or our region?

Environmental Law and Policy

- Uzbekistan:
- Constitution
- Law on the protection of nature (Part VIII, Articles 33-34, Part X, Articles 41-44)
- Government decrees
- National Committee on Ecology and Climate Change of the Republic of Uzbekistan
- Globally:
- UN, Greenpeace

Applying Environmental Laws – Clean Enough?

- When the environment is acceptably clean?
 1. Comparative criterion
 2. Normalcy criterion
 3. Optimal pollution reduction criterion
 4. Maximum protection criterion
 5. Demonstrable harm criterion
 6. Degree of harm criterion

Life Cycle Analysis (LCA)

- LCA is a cradle-to-grave analysis of the environmental impact of a product or a process.
- It covers the life history of a product or process from the extraction of raw materials from the earth, through manufacture and use, to its final disposal
- “Green” products
- Limitations?

BOX 7.1 Four Phases of Life Cycle Analysis

1. *Goal and Scope.* Defining the product or process, the context of the assessment, the boundaries of the analysis (geographical area, temporal boundaries of the study, and boundaries between this analysis and related life cycles of other systems), and the environmental effects.
2. *Inventory Analysis.* Relevant inputs and outputs of a product or process in terms of the energy, water, and materials used and identification and quantification of releases.
3. *Impact Assessment.* Identification and quantification of the most significant environmental impacts associated with the product, including resource use, human health and ecological consequences, and greenhouse gas emissions.
4. *Interpretation.* Evaluation of the results of the first three phases, along with evaluations of the assumptions made and the degree of uncertainty assumed. The best product or process is then selected.

Sustainability: the Environment Versus Human Development

- What is sustainability?

Sustainability: the Environment Versus Human Development

- “To sustain” is to keep in being or to cause to continue in a certain state.
- “the process of applying natural, human, and economic resources to enhance the safety, welfare, and quality of life for all of society while maintaining the availability of remaining natural resources.”
- “development that meets the needs of the present without compromising the ability of future generations to meet their own needs.” (Brundtland Report)

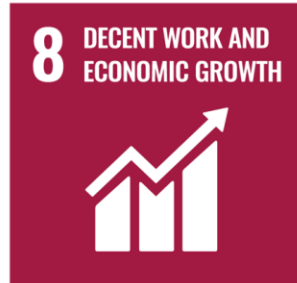
BOX 7.2 Five Goals for Sustainable Development According to the Brundtland Report

1. Economic growth
2. Fair distribution of resources to sustain economic development
3. More democratic political systems
4. Adoption of lifestyles that are more compatible with living within the planet's ecological means
5. Population levels that are more compatible with the planet's ecological means.

Any issues?



SUSTAINABLE DEVELOPMENT GOALS



The Moral Case for Sustainable Development

- Utilitarian standpoint – future generations of humans (and even animals) should be included with the present generation in the audience over which utility is maximized
- Problems?
- RP standpoint – we have an obligation to respect equally the rights of all people (both present and future generations)
- Problems?
- Virtue ethics – the virtues of care and respect for nature. Care should extend to all members of the present generation as well as future generations, especially care for the poor and disadvantaged.

The Moral Case for Sustainable Development

- Environmental or Social Collapse? – Utilitarian perspectives
- What will happen if the human community fails to adopt some form of sustainable development?
- Thomas Malthus (1798) – populations grow geometrically, whereas food increases arithmetically or even linearly
- Garrett Hardin – concept of the “tragedy of the commons”
- Resource collapse or near-collapse has already occurred
- Precautionary principle – where there is risk of irreversible or serious damage to the environment, lack of full scientific certainty shall not constitute a reason to postpone preventive or ameliorative action.

Sustainable Development and Engineering Practice

- Challenges of Implementation
- Pure ideal of sustainability?
- Cradle to Grave: production and recycling
- Problems?
- Cradle to Cradle - follow natural processes, many of which are circular. Nature recycles everything
- Humans don't have a pollution problem, they have a design problem

Business and Sustainable Development

■ Three Industry Attitudes Toward the Environment:

- The subminimal attitude
- The minimalist or compliance attitude
- The progressive attitude

Cultivating the Progressive Attitude

- Good for business usually means good for enhancing a firm's public image or good for the bottom line or both

BOX 7.4 Stages in the Development of an Environment-Friendly Firm

1. Viewing compliance as a challenge to innovate and complying with the most stringent rules, so as to be ahead of other firms when the more stringent rules are enforced and believing that this approach can give them a market advantage.
2. Designing the firm's own products and services to be more sustainable.
3. Requiring suppliers to make their operations more sustainable by methods such as developing more fuel-efficient vehicles and machines.
4. Turning waste and pollutants into valuable products that can be sold for profit.
5. Questioning the implicit assumptions behind products and services and thereby thinking "outside the box." (Can we develop waterless detergents?)

Cultivating the Virtue of Respect for Nature

- The Healing/Restoring Aspect of Nature:
- Recent research has confirmed that the effects of the environment on human psychology are remarkable
- The Emotional Effects of Experiencing the Natural World as Transcending Human Interests:
- Appreciating the natural world as transcending human interests can also contribute to human well-being and to the development of the virtue of respect for nature